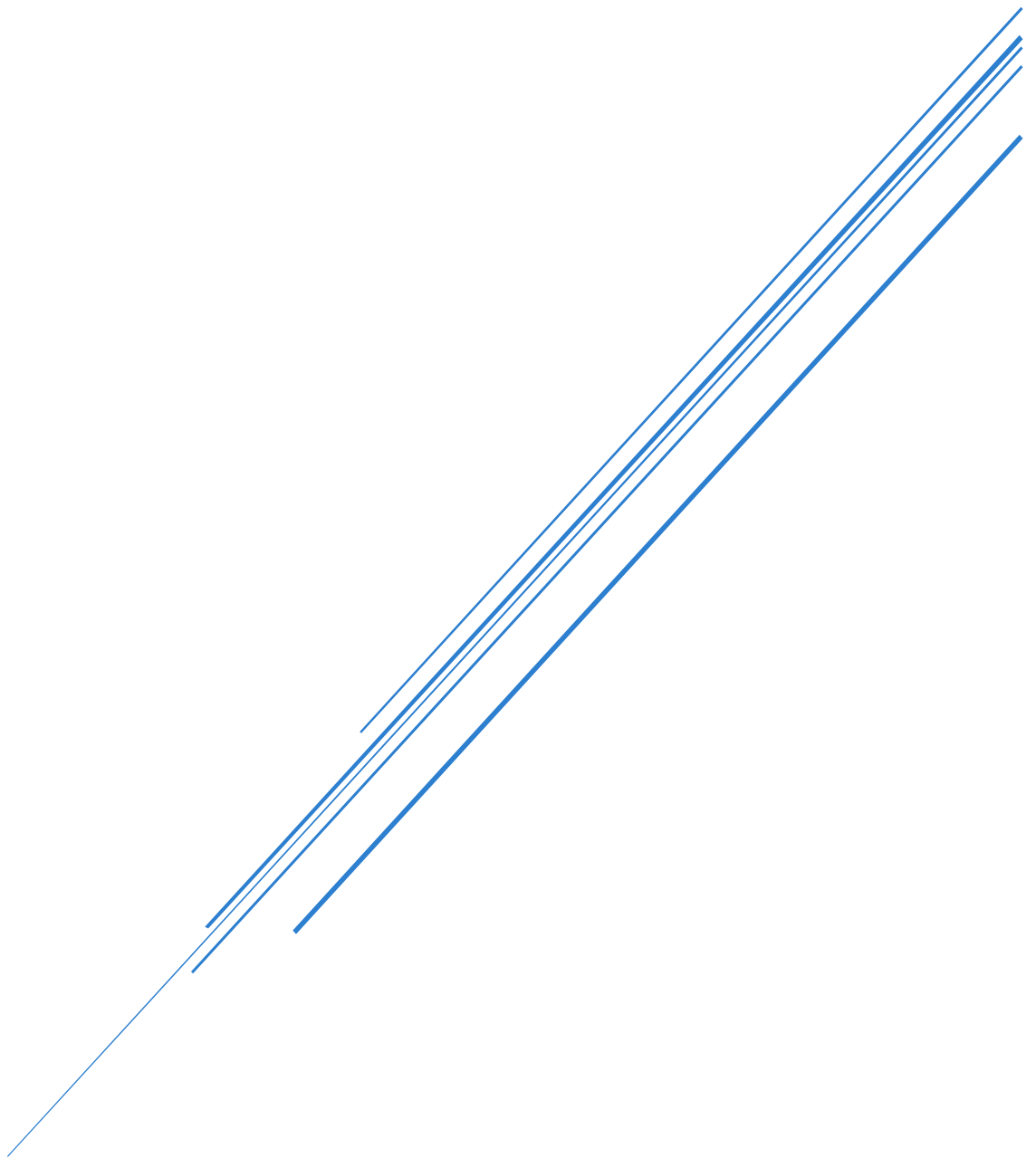


POWER BI

Assignment 1 – Data Transformation & Modelling



Power BI Assignment 1 – Data Transformation & Data Modelling

Dataset Description:

This assignment has three datasets

- **List of orders**

Attributes: Order ID, Order Date, Customer Name, City, State, Category, Amount, Profit

- **Order Details**

Attributes: Order ID, Category, Sub-Category, Quantity, Amount, Profit

- **Sales Target**

Attributes: Category, Month of Order Date, Target

Data Import & Transformation

1. Data Import:

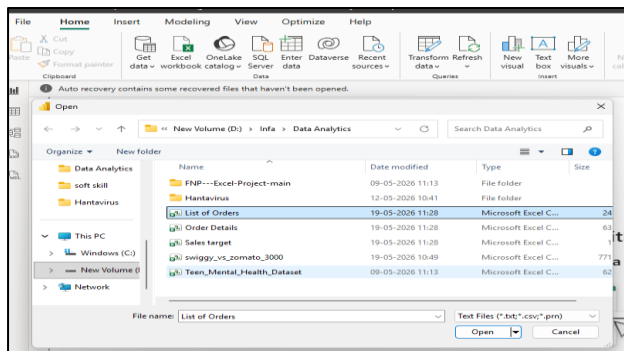
- Import List of Orders.csv into Power BI.
- Open the list of orders in the Power Query Editor using the Transform Data option.
- Import Order Details.csv and Sales Target.csv into Power Query Editor.

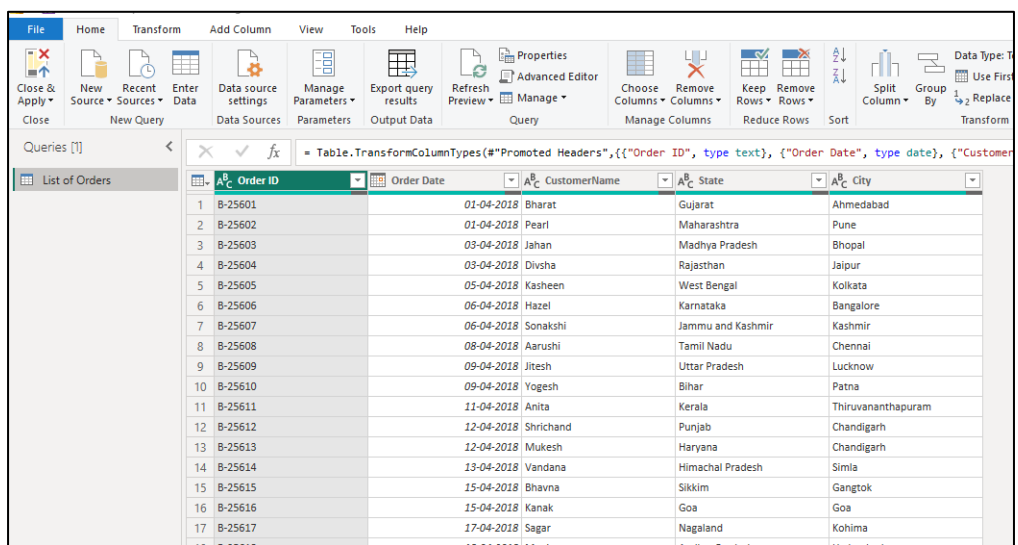
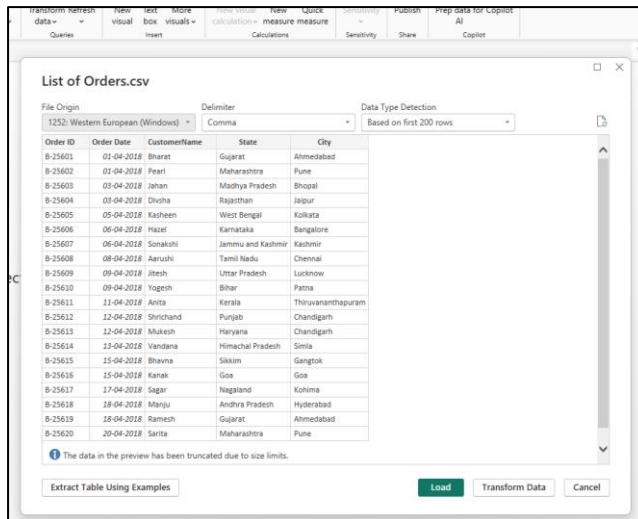
Initially, I checked the file format of the three datasets which is for.csv file. Therefore, I followed the below steps to import the files.

Loading of List of Order Dataset:

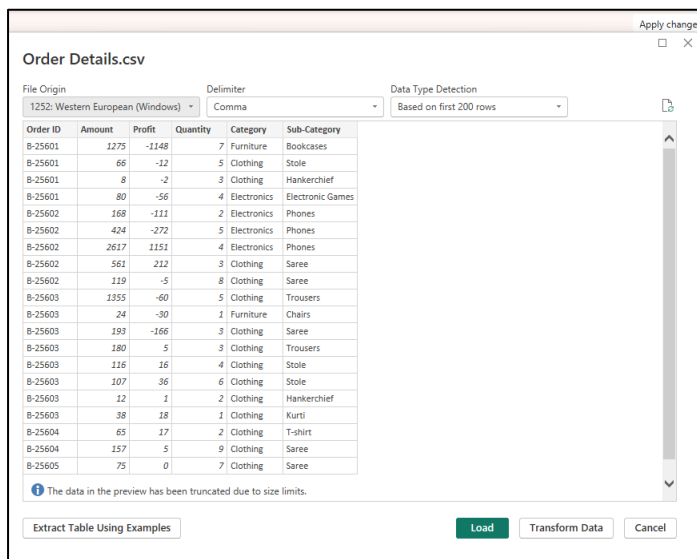
1. Open Power BI – In Home -> Goto Get Data in data tab -> Select text/CSV
2. In the open dialogue box -> Select the file you want to load (i.e list of orders) -> Click Open -> Select Transform Data

Followed the above 2 steps for the remaining two datasets as well





Loading of Order Details Dataset:



File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Queries [2]

Order Details

Order ID Amount Profit Quantity Category Sub-Category

Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	1275	-1148		Furniture	Bookcases
2	66	-12		Clothing	Stole
3	8	-2		Clothing	Hankkerchief
4	80	-56		Electronics	Electronic Games
5	168	-111		Electronics	Phones
6	424	-272		Electronics	Phones
7	2617	1151		Electronics	Phones
8	561	212		Clothing	Saree
9	119	-5		Clothing	Saree
10	1355	-60		Clothing	Trousers
11	24	-30		Furniture	Chairs
12	193	-166		Clothing	Saree
13	180	5		Clothing	Trousers
14	116	16		Clothing	Stole
15	107	26		Clothing	Stole

Loading of Sales Target Dataset:

Queries Insert Calculations Sensitivity Share Copilot

Apply changes

Sales target.csv

File Origin Delimiter Data Type Detection

1252: Western European (Windows) Comma Based on first 200 rows

Month of Order Date	Category	Target
01-04-2018	Furniture	10400
01-05-2018	Furniture	10500
01-06-2018	Furniture	10600
01-07-2018	Furniture	10800
01-08-2018	Furniture	10900
01-09-2018	Furniture	11000
01-10-2018	Furniture	11100
01-11-2018	Furniture	11300
01-12-2018	Furniture	11400
01-01-2019	Furniture	11500
01-02-2019	Furniture	11600
01-03-2019	Furniture	11800
01-04-2019	Clothing	12000
01-05-2019	Clothing	12000
01-06-2019	Clothing	12000
01-07-2019	Clothing	14000
01-08-2019	Clothing	14000
01-09-2019	Clothing	14000
01-10-2019	Clothing	16000
01-11-2019	Clothing	16000

The data in the preview has been truncated due to size limits.

Extract Table Using Examples Load Transform Data Cancel

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Queries [3]

Order Details

Sales target

Month of Order Date Category Target

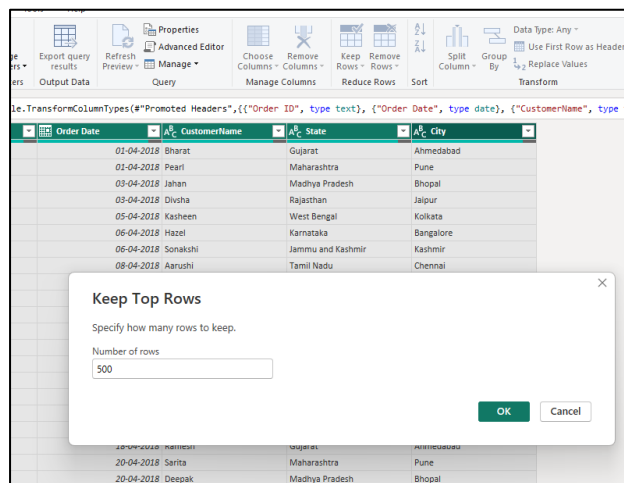
Month of Order Date	Category	Target
01-04-2018	Furniture	10400
01-05-2018	Furniture	10500
01-06-2018	Furniture	10600
01-07-2018	Furniture	10800
01-08-2018	Furniture	10900
01-09-2018	Furniture	11000
01-10-2018	Furniture	11100
01-11-2018	Furniture	11300
01-12-2018	Furniture	11400
01-01-2019	Furniture	11500
01-02-2019	Furniture	11600

2. Row Limiting & Data Types:

○ Restrict the List of Orders table to the first 500 rows.

Keep Top Rows:

1. Open Power Query Editor:
 - Home → Transform data
2. Select list of orders table.
3. Go to:
 - Home tab → Keep Rows → Keep Top Rows
4. Enter:
 - 500
5. Click OK



Result:

Order ID	Order Date	Customer	City
1 B-25601	01-04-2018	Bharat	
2 B-25602	01-04-2018	Pearl	
3 B-25603	03-04-2018	Jahan	
4 B-25604	03-04-2018	Divsha	
5 B-25605	05-04-2018	Kasheen	
6 B-25606	06-04-2018	Hazel	
7 B-25607	06-04-2018	Sonakshi	
8 B-25608	08-04-2018	Aanushi	
9 B-25609	09-04-2018	Itish	
10 B-25610	09-04-2018	Yogesh	
11 B-25611	11-04-2018	Anita	
12 B-25612	12-04-2018	Shrichand	
13 B-25613	12-04-2018	Mukesh	
14 B-25614	12-04-2018	Vandana	
15 B-25615	15-04-2018	Bheema	
16 B-25616	15-04-2018	Kanak	
17 B-25617	17-04-2018	Sagar	
18 B-25618	18-04-2018	Manju	
19 B-25619	18-04-2018	Ramesh	
20 B-25620	20-04-2018	Sarita	
21 B-25621	20-04-2018	Deepak	
22 B-25622	22-04-2018	Monisha	
23 B-25623	22-04-2018	Atharv	
24 B-25624	22-04-2018	Vini	
25 B-25625	23-04-2018	Pinky	
26 B-25626	23-04-2018	Bhishm	
27 B-25627	23-04-2018	Hitesh	
28 B-25628	24-04-2018	Pooja	
29 B-25629	24-04-2018	Hemant	
30 B-25630	24-04-2018	Sahil	
31 B-25631	24-04-2018	Ritu	
32 B-25632	25-04-2018	Manish	
33 B-25633	26-04-2018	Armit	
34 B-25634	26-04-2018	Sampy	
35 B-25635	26-04-2018	Nidhi	
36 B-25636	26-04-2018	Nishi	
37 B-25637	26-04-2018	Ashvi	
38 B-25638	26-04-2018	Parth	

○ Set the Order Date column to the Date data type.

Select the Order date column -> In the left corner of the header date column we need to click and select the datatype as date as seen below,

The screenshot shows the Power BI Desktop interface with the 'Transform' tab selected. The 'Order Date' column is highlighted, and the data type is set to 'Date'.

Order ID	Order Date	CustomerName	State	City
1	01-04-2018	Bharat	Gujarat	Ahmedabad
2	01-04-2018	Pearl	Maharashtra	Pune
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	03-04-2018	Divsha	Rajasthan	Jaipur
5	05-04-2018	Kasheen	West Bengal	Kolkata
6	06-04-2018	Hazel	Karnataka	Bangalore
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	09-04-2018	Yogesh	Bihar	Patna
11	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	12-04-2018	Shrichand	Punjab	Chandigarh
13	12-04-2018	Mukesh	Haryana	Chandigarh

○ Change the Amount and Target columns to fixed decimal numbers.

In the order details dataset table, select the column "amount" -> In the left corner of the header amount column we need to click and select the datatype as fixed decimal number as seen below,

The screenshot shows the Power BI Desktop interface with the 'Transform' tab selected. The 'Amount' column is highlighted, and the data type is set to 'Fixed Decimal Number'.

Order ID	Amount	Profit	Quantity	Category
1	1275	-1148	7	Furniture
2	66	-12	5	Clothing
3	8	-2	3	Clothing
4	80	-56	4	Electronics
5	168	-111	2	Electronics
6	424	-272	5	Electronics
7	2617	1151	4	Electronics
8	561	212	3	Clothing
9	119	-5	8	Clothing
10	1355	-60	5	Clothing
11	24	-30	1	Furniture
12	193	-166	3	Clothing
13	180	5	3	Clothing
14	116	16	4	Clothing
15	107	26	6	Clothing

= Table.TransformColumnTypes("#Promoted Headers",{{"Amount"			
	A ^B Order ID	\$ Amount	A ^B Profit
1	B-25601	1,275.00	-1148.00
2	B-25601	66.00	-12.00
3	B-25601	8.00	-2.00
4	B-25601	80.00	-56.00
5	B-25602	168.00	-111.00
6	B-25602	424.00	-272.00
7	B-25602	2,617.00	1151.00
8	B-25602	561.00	212.00
9	B-25602	119.00	-5.00
10	B-25603	1,355.00	-60.00
11	B-25603	24.00	-30.00
12	B-25603	193.00	-166.00
13	B-25603	180.00	5.00
14	B-25603	116.00	16.00
15	B-25603	107.00	36.00
16	B-25603	12.00	1.00

In the sales target dataset table, select the column “target” -> In the left corner of the header amount column we need to click and select the datatype as fixed decimal number as seen below,

= Table.TransformColumnTypes("#Changed Type",{{"Target", Currency.Type}}			
	Month of Order Date	A ^B Category	\$ Target
1	01-04-2018	Furniture	10,400.00
2	01-05-2018	Furniture	10,500.00
3	01-06-2018	Furniture	10,600.00
4	01-07-2018	Furniture	10,800.00
5	01-08-2018	Furniture	10,900.00
6	01-09-2018	Furniture	11,000.00
7	01-10-2018	Furniture	11,100.00
8	01-11-2018	Furniture	11,300.00
9	01-12-2018	Furniture	11,400.00
10	01-01-2019	Furniture	11,500.00
11	01-02-2019	Furniture	11,600.00
12	01-03-2019	Furniture	11,800.00
13	01-04-2018	Clothing	12,000.00
14	01-05-2018	Clothing	12,000.00

3. Text Formatting:

○ Format the Customer Name column to proper case to ensure consistent capitalization.

1. Select the “Customer Name” column in list of orders table
2. Select the “Format” option in text column tab
3. Choose Capitalize Each word

Table.TransformColumnTypes("#Kept First Rows",{{"Order Date", type date}})					
Order ID	Order Date	CustomerName	State	City	
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh

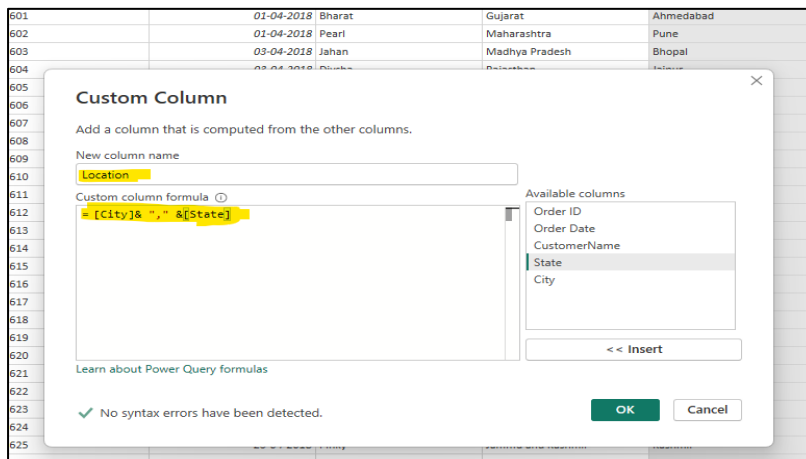
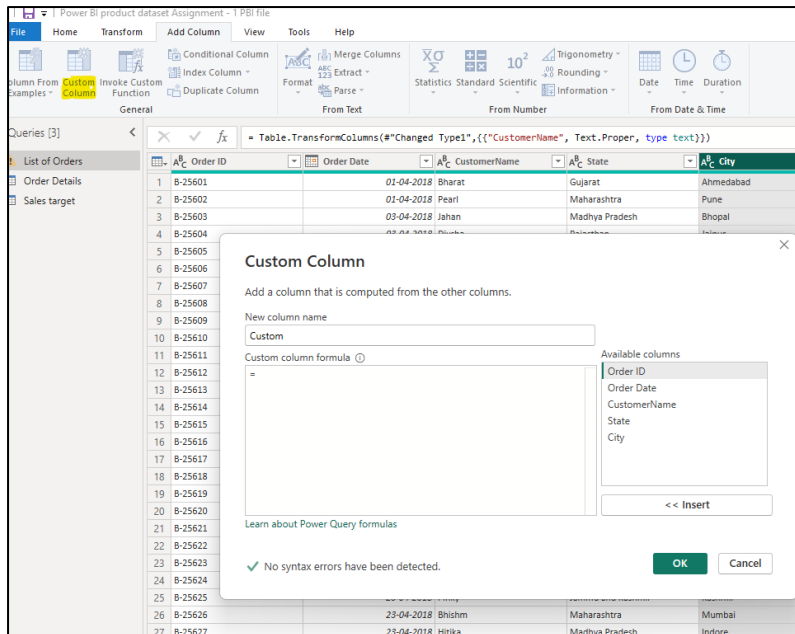
Table.TransformColumns("#Changed Type1",{{"CustomerName", Text.Proper, type text}})					
Order ID	Order Date	CustomerName	State	City	
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima

4. Column Creation:

○ Merge the City and State columns to create a new column named Location in the format: City, State

In the list of orders table -> Select any column and follow the below steps.

1. In Add Column tab -> select Custom Column
2. In custom column dialog box -> give "Location" as a new column name
3. In custom column formula, provide the below formula to add merge the two columns ("&" used to merge two text and comma used to separate the text) and click OK
4. Formula Used : = [City]& "," &[State]



Result:

Number	From Date & Time		
ation", each [City]& ", " &[State])			
ne	A ^B _C State	A ^B _C City	ABC ₁₂₃ Location
	Gujarat	Ahmedabad	Ahmedabad,Gujarat
	Maharashtra	Pune	Pune,Maharashtra
	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
	Rajasthan	Jaipur	Jaipur,Rajasthan
	West Bengal	Kolkata	Kolkata,West Bengal
	Karnataka	Bangalore	Bangalore,Karnataka
	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
	Tamil Nadu	Chennai	Chennai,Tamil Nadu
	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
	Bihar	Patna	Patna,Bihar
	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
	Punjab	Chandigarh	Chandigarh,Punjab
	Haryana	Chandigarh	Chandigarh,Haryana
	Himachal Pradesh	Simla	Simla,Himachal Pradesh
	Sikkim	Gangtok	Gangtok,Sikkim
	Goa	Goa	Goa,Goa

- Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

In the order details table, first check the data types of the columns profit and Amount and changed it according to the headers.

Select any column and follow the below steps.

1. In Add Column tab -> select Custom Column
2. In custom column dialog box -> give "Profit Margin" as a new column name
3. In custom column formula, provide the below formula and click OK
4. Formula Used : $=[Profit]/[Amount]$
5. Then, change the datatype of profit margin column to %

Table.TransformColumnTypes(#"Promoted Headers",{{"Profit", Int64.Type}, {"Amount", Currency.Type}})

	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601	1,275.00	-1148	7	Furniture	Bookcases
2	B-25601	66.00	-12	5	Clothing	Stole
3	B-25601	8.00	-2	3	Clothing	Hankerchi
4	B-25601	80.00	-56	4	Electronics	Electronic
5	B-25602	168.00	-111	2	Electronics	Phones
6	B-25602	424.00	-272	5	Electronics	Phones
7	B-25602	2,617.00	1151	4	Electronics	Phones
8	B-25602	561.00	212	3	Clothing	Saree
9	B-25602	119.00	-5	8	Clothing	Saree
10	B-25603	1,355.00	-60	5	Clothing	Trousers
11	B-25603	24.00	-30	1	Furniture	Chairs
12	B-25603	193.00	-166	3	Clothing	Saree
13	B-25603	180.00	5	3	Clothing	Trousers
14	B-25603	116.00	16	4	Clothing	Stole
15	B-25603	107.00	36	6	Clothing	Stole
16	B-25603	12.00	1	2	Clothing	Hankerchi
17	B-25603	38.00	18	1	Clothing	Kurti

Table.TransformColumnTypes(#"Promoted Headers",{{"Amount", Currency.Type}})

ID	Amount	Profit	Quantity	Category	Sub-Category
	1,275.00	-1148.00	7	Furniture	Bookcases
	66.00	-12.00	5	Clothing	Stole
	8.00	-2.00	3	Clothing	Hankerchi
	80.00	-56.00	4	Electronics	Electronic
					Phones
					Phones
					Phones
					Saree
					Saree
					Trousers
					Chairs
					Saree
					Trousers
					Stole
					Stole
					Hankerchi
					Kurti
					T-shirt
					Saree
					Saree
					Shirt
					Legging
					Tables
					Chairs
					Hankerchi
					Printers
					Electronics

Custom Column

Add a column that is computed from the other columns.

New column name: Profit Margin

Custom column formula: $=[Profit]/[Amount]$

Available columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

m Date & Time		
ABC	ABC	Profit Margin
Category	Sub-Category	
Furniture	Bookcases	-0.900392157
Clothing	Stole	-0.181818182
Clothing	Hankerchief	-0.25
Electronics	Electronic Games	-0.7
Electronics	Phones	-0.660714286
Electronics	Phones	-0.641509434
Electronics	Phones	0.439816584
Clothing	Saree	0.377896613
Clothing	Saree	-0.042016807
Clothing	Trousers	-0.044280443
Furniture	Chairs	-1.25
Clothing	Saree	-0.860103627
Clothing	Trousers	0.027777778
Clothing	Stole	0.137931034
Clothing	Stole	0.336448598

Result:

ABC	ABC	% Profit Margin
Sub-Category		
Bookcases		-90.04%
Stole		-18.18%
Hankerchief		-25.00%
Electronic Games		-70.00%
Phones		-66.07%
Phones		-64.15%
Phones		43.98%
Saree		37.79%
Saree		-4.20%
Trousers		-4.43%
Chairs		-125.00%
Saree		-86.01%
Trousers		2.78%
Stole		13.79%
Stole		33.64%
Hankerchief		8.33%
Kurti		47.37%

5. Conditional Column:

○ Add a conditional column named Profit Status based on:

- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit

In the order details table,

Select any column and follow the below steps.

1. In Add Column tab -> select Conditional Column
2. In add Conditional Column dialog box -> give "Profit Status" as a new column name

3. Then, provide the below formula (add additional Clause if needed) -> click OK

Formula Used: Profit > 0 → "Profit"

Profit < 0 → "Loss"

Profit = 0 → "Break-even"

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name: Profit Status

	Column Name	Operator	Value	Output
If	Profit	is less than	0	Loss
Else If	Profit	equals	0	Break-Even

Buttons: Add Clause, OK, Cancel

Result:

From Number		From Date & Time			
margin to percentage", "Profit Status", each if [Profit] < 0 then "Loss" else if [Profit] = 0 then "Break-Even" else "Profit")					
123 Profit	A _C Quantity	A _C Category	A _C Sub-Category	% Profit Margin	A ₁₂₃ Profit Status
-1148	7	Furniture	Bookcases	-90.04%	Loss
-12	5	Clothing	Stole	-18.18%	Loss
-2	3	Clothing	Hankerchief	-25.00%	Loss
-56	4	Electronics	Electronic Games	-70.00%	Loss
-111	2	Electronics	Phones	-66.07%	Loss
-272	5	Electronics	Phones	-64.15%	Loss
1151	4	Electronics	Phones	43.98%	Profit
212	3	Clothing	Saree	37.79%	Profit
-5	8	Clothing	Saree	-4.20%	Loss
-60	5	Clothing	Trousers	-4.43%	Loss
-30	1	Furniture	Chairs	-125.00%	Loss
-166	3	Clothing	Saree	-86.01%	Loss
5	3	Clothing	Trousers	2.78%	Profit
16	4	Clothing	Stole	13.79%	Profit
36	6	Clothing	Stole	33.64%	Profit
1	2	Clothing	Hankerchief	8.33%	Profit
18	1	Clothing	Kurti	47.37%	Profit
17	2	Clothing	T-shirt	26.15%	Profit

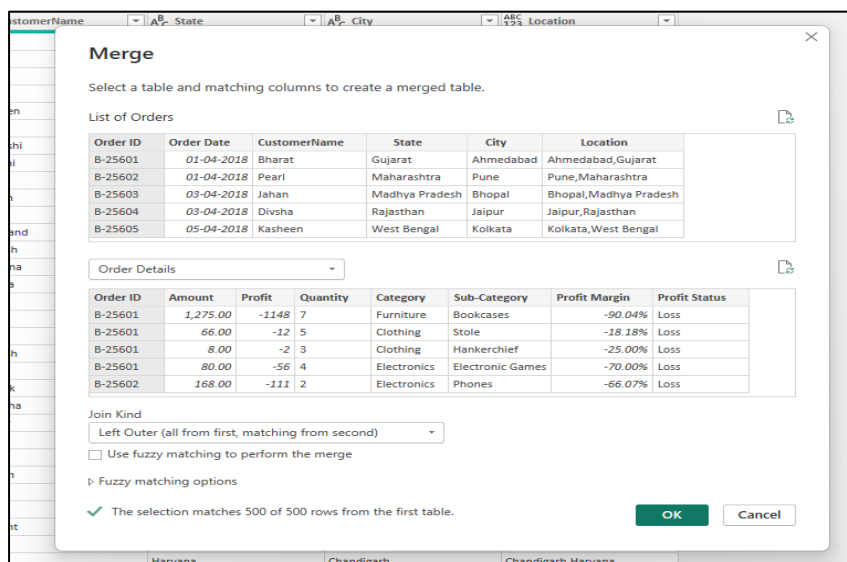
6. Merging Data (Joins)

○ Merge the List of Orders and Order Details tables using Order ID.

To merge the List of Orders and Order Details tables using Order ID in Power BI Power Query:

Steps to Merge Tables

1. Open Power Query Editor
 - Home → Transform Data
2. Select the table:
 - List of Orders
3. Go to:
 - Home → Merge Queries
4. In the Merge dialog:
 - First table: List of Orders
 - Second table: Order Details
 - Select the Order ID column in both tables
5. Choose the Join Type:
 - Usually: Left Outer (all from first, matching from second)
6. Click OK



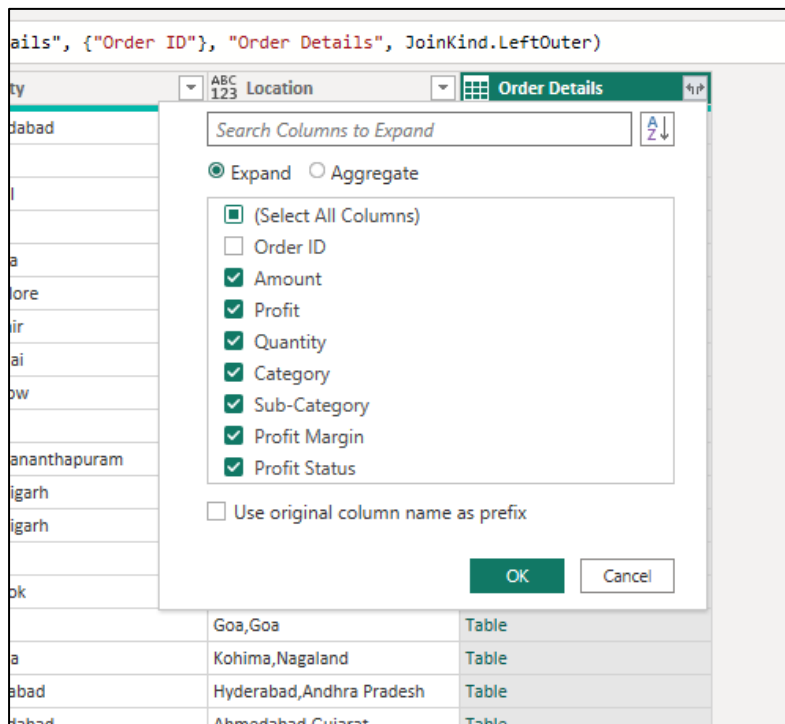
Expanding the Merged Table After merging:

1. A new column appears containing table values.
2. Click the expand icon (↔) next to the column.
3. Select the columns you want from Order Details

Example:

- Amount
- Profit
- Quantity
- Category
- Sub-Category

4. Uncheck:
 - Use original column name as prefix

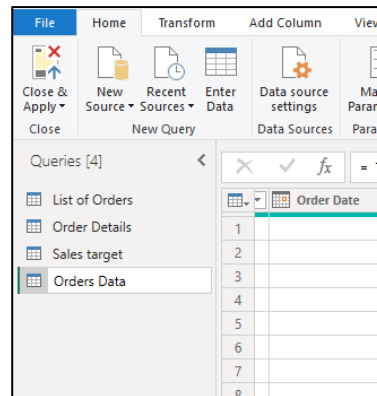
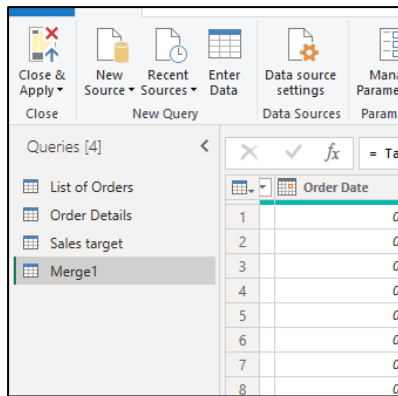


Location	\$ Amount	Profit	Quantity
Ahmedabad, Gujarat	1,275.00	-1148	7
Ahmedabad, Gujarat	66.00	-12	5
Ahmedabad, Gujarat	8.00	-2	3
Ahmedabad, Gujarat	80.00	-56	4
Pune, Maharashtra	168.00	-111	2
Pune, Maharashtra	424.00	-272	5
Pune, Maharashtra	2,617.00	1151	4
Pune, Maharashtra	561.00	212	3
Pune, Maharashtra	119.00	-5	8
Bhopal, Madhya Pradesh	1,355.00	-60	5
Bhopal, Madhya Pradesh	24.00	-30	1
Bhopal, Madhya Pradesh	193.00	-166	3
Bhopal, Madhya Pradesh	180.00	5	3
Bhopal, Madhya Pradesh	116.00	16	4
Bhopal, Madhya Pradesh	107.00	36	6
Bhopal, Madhya Pradesh	12.00	1	2
Bhopal, Madhya Pradesh	38.00	18	1

○ Name the merged output table as Orders Data.

Rename the Output Table In the left Queries pane:

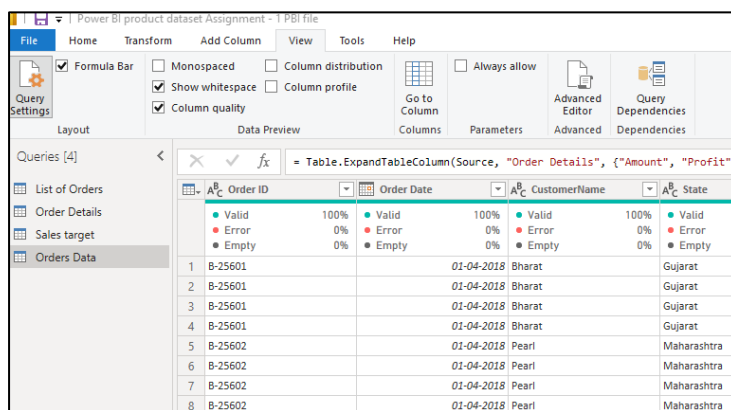
- Right-click the merged query
- Rename it to: Orders Data



7. Handling Missing Data & Duplicate Data

- Identify missing values across all tables.

Missing values were identified using column filters in Power Query Editor and empty/blank cells by use of column quality in view tab



- Define and apply appropriate strategies such as:

- Replacing missing values
- Removing records
- Retaining nulls with justification

In the given dataset, I couldn't find any missing values and everything is good.

Regarding the Sales target table, I could find a column named "Month of order date", however it has complete date instead of month. Therefore, I am adding new column named as "month of order" by converting the original date column -> date -> month -> Name of month. Then renaming the date column as "Order Date" and new column as "Month of Order date"

File	Home	Transform	Add Column	View	Tools	Help
Column From Examples	Custom Column	Invoke Custom Function	Conditional Column	Index Column	Duplicate Column	
General			Format		From Text	
Queries [8]			= Table.TransformColumnTypes(#"Changed Type",{{"Target", Currency.Type}})			
List of Orders	Order Details	Sales target	Month of Order Date	Category	Target	
1	01-04-2018	Furniture		10,400.00		
2	01-05-2018	Furniture		10,500.00		

From Text		From Number		From Date & Time	
= Table.AddColumn(#"datatype num changed to fixed decimal num in target column", "Month Name")					
Month of Order Date	Category	Target	Month Name		
1	01-04-2018	Furniture	10,400.00	April	
2	01-05-2018	Furniture	10,500.00	May	
3	01-06-2018	Furniture	10,600.00	June	
4	01-07-2018	Furniture	10,800.00	July	
5	01-08-2018	Furniture	10,900.00	August	
6	01-09-2018	Furniture	11,000.00	September	
7	01-10-2018	Furniture	11,100.00	October	
8	01-11-2018	Furniture	11,300.00	November	
9	01-12-2018	Furniture	11,400.00	December	
10	01-01-2019	Furniture	11,500.00	January	
11	01-02-2019	Furniture	11,600.00	February	
12	01-03-2019	Furniture	11,800.00	March	
13	01-04-2018	Clothing	12,000.00	April	
14	01-05-2018	Clothing	12,000.00	May	
15	01-06-2018	Clothing	12,000.00	June	

● Identify duplicate rows and define a suitable handling strategy based on business logic.

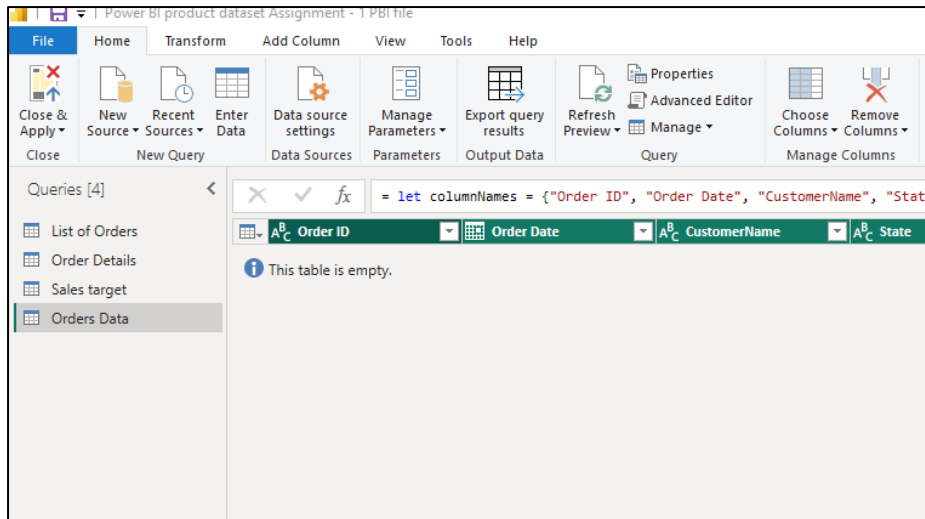
Before removing, review whether duplicates are true duplicates or valid repeated transactions.

However, in this case, Duplicate Order IDs were retained because a single order can contain multiple products.

Remove Exact Duplicates

Used when:

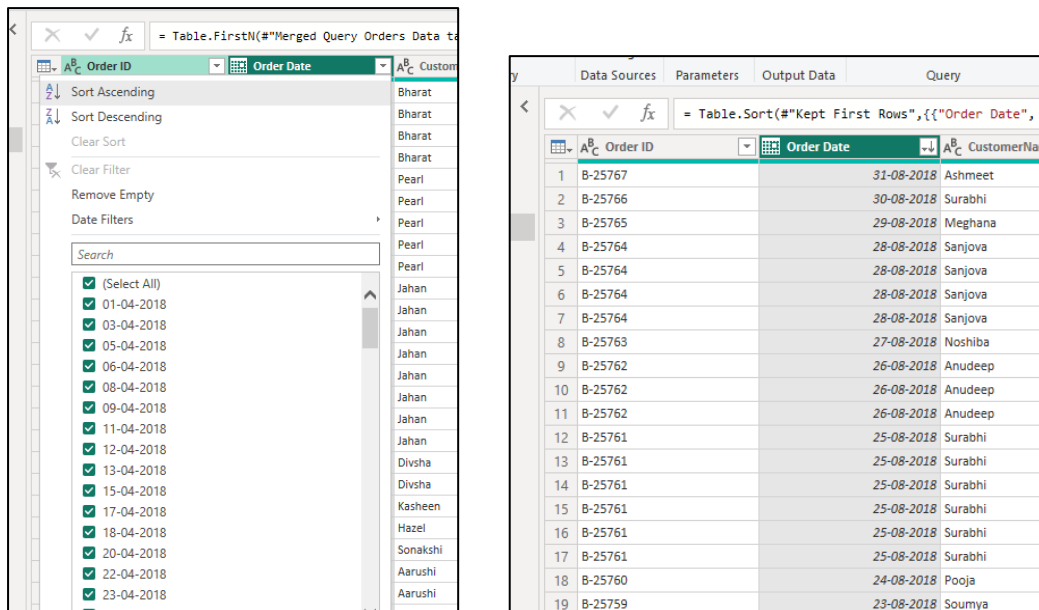
- Entire rows are repeated accidentally.
- However, in this no repeated rows are found
- We can find this by **Keep Rows** → **Keep Duplicates** - This shows only duplicate rows.
- However, there are no duplicates found per below in the given dataset



8. Sorting and Filtering Data

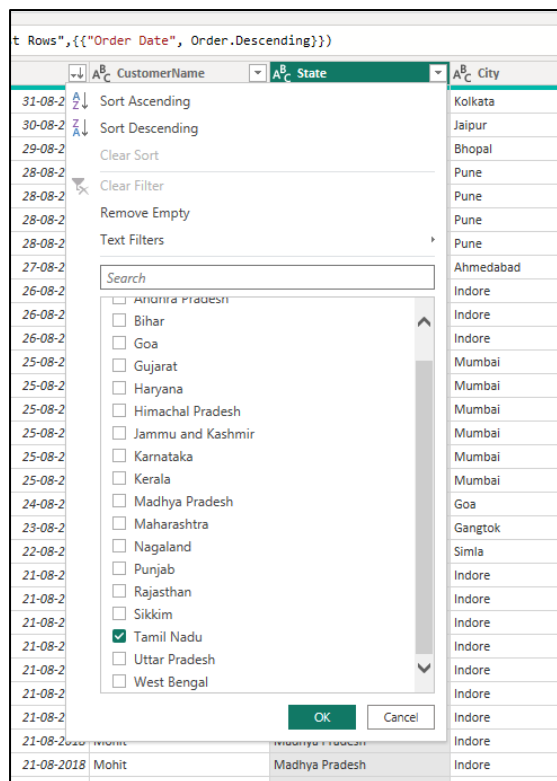
- Sort records in the Orders Data table by Order Date in descending order.

In the Orders Data table, select the column Order date -> Click the right corner drop down -> Select Sort Descending



- Apply filters to analyse orders from a specific state (e.g., Tamil Nadu).

In the orders data table, Find the State column -> Click the filter dropdown -> Uncheck Select All -> Check “ Tamil Nadu”



Order Details to Descending Order", each ([State] = "Tamil Nadu"))

CustomerName	State	City
rabhi	Tamil Nadu	Chennai
nisha	Tamil Nadu	Chennai
nisha	Tamil Nadu	Chennai
nisha	Tamil Nadu	Chennai
nisha	Tamil Nadu	Chennai
nisha	Tamil Nadu	Chennai
nisha	Tamil Nadu	Chennai
rushi	Tamil Nadu	Chennai
rushi	Tamil Nadu	Chennai
rushi	Tamil Nadu	Chennai
rushi	Tamil Nadu	Chennai

9. Grouping and Aggregating Data

● Duplicate the Order Details table and perform the following:

- Count of Order ID
- Average Profit by Category
- Total Amount by Sub-Category

Step 1: Duplicate Table

1. Open **Power Query Editor**
2. In the left Queries pane:
 - Right-click Order Details
 - Select: Duplicate
 - Rename the duplicated table: Order Details Summary

Performing Aggregations

A. Count of Order ID

Steps:

1. Select Order Details Summary

2. Go to:
 - Home → Group By
 3. Configure:
 - Group By: Order ID
 - New column name: Order Count
 - Operation: Count Rows
 4. Click OK
-

B. Average Profit by Category

Steps:

1. Duplicate original Order Details again
 2. Rename:
 - Avg Profit by Category
 3. Go to:
 - Home → Group By
 4. Configure:
 - Group By: Category
 - New column: Average Profit
 - Operation: Average
 - Column: Profit
 5. Click OK
-

C. Total Amount by Sub-Category

Steps:

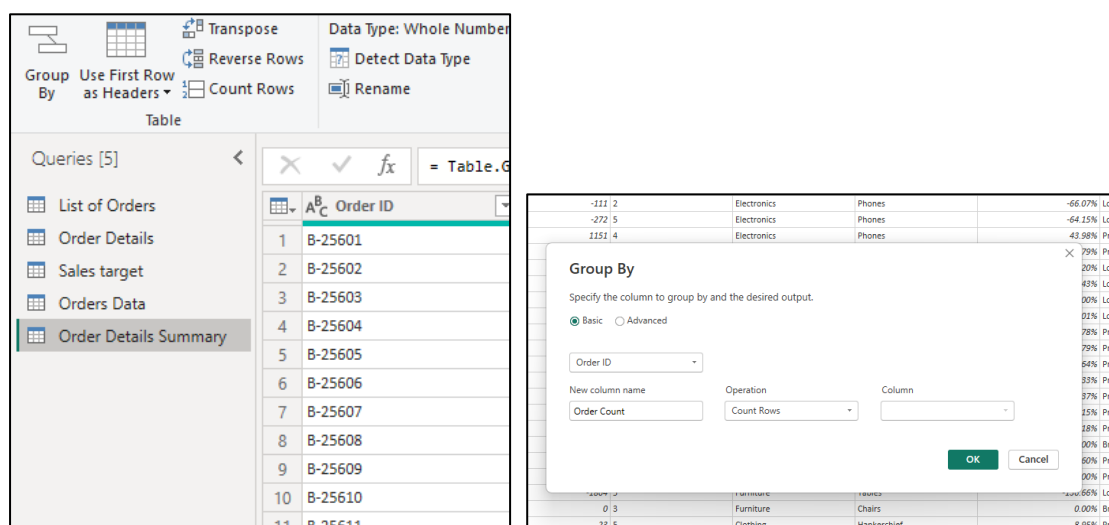
1. Duplicate original Order Details
2. Rename:
 - Total Amount by SubCategory
3. Go to:

- Home → Group By

4. Configure:

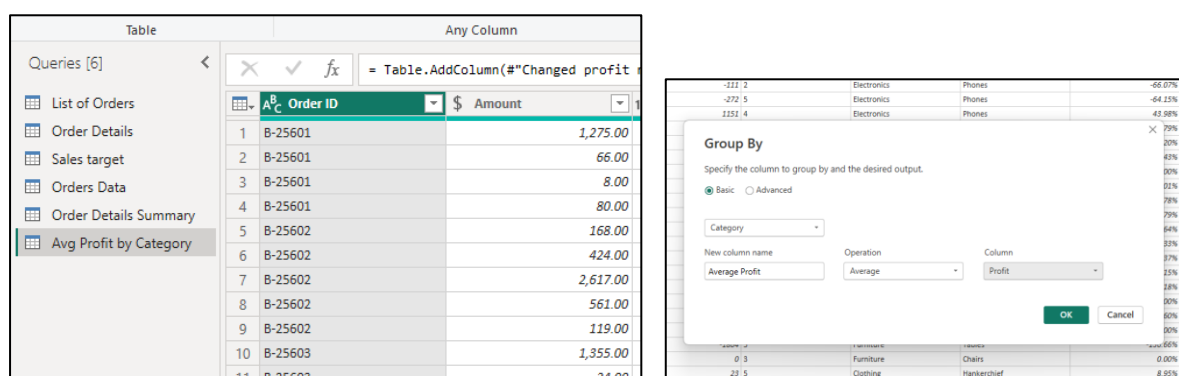
- Group By: Sub-Category
- New column: Total Amount
- Operation: Sum
- Column: Amount

5. Click OK



The screenshot shows the Power BI Desktop interface. On the left, the 'Queries' pane lists 'List of Orders', 'Order Details', 'Sales target', 'Orders Data', and 'Order Details Summary'. The main area displays a table with columns 'Order ID' and 'Sub-Category'. The 'Group By' dialog box is open, showing the 'Basic' tab. The 'Group By' field is set to 'Order ID', and the 'Operation' is set to 'Count Rows'. The 'New column name' is 'Order Count'. The 'Column' field is empty. The 'OK' button is highlighted.

Table	Any Column
Queries [6]	= Table.Group("#Added Conditional Column as P
List of Orders	
Order Details	
Sales target	
Orders Data	
Order Details Summary	
Order ID	Order Count
1 B-25601	4
2 B-25602	5
3 B-25603	8
4 B-25604	2



The screenshot shows the Power BI Desktop interface. On the left, the 'Queries' pane lists 'List of Orders', 'Order Details', 'Sales target', 'Orders Data', 'Order Details Summary', and 'Avg Profit by Category'. The main area displays a table with columns 'Order ID' and 'Amount'. The 'Group By' dialog box is open, showing the 'Basic' tab. The 'Group By' field is set to 'Category', and the 'Operation' is set to 'Average'. The 'New column name' is 'Average Profit'. The 'Column' field is set to 'Profit'. The 'OK' button is highlighted.

Group By Use First Row as Headers Count Rows Rename Pivot Column Convert to List

Table Any Column

Queries [6]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Order Details Summary
- Avg Profit by Category

fx = Table.Group(#"Added Conditional Column", "Category", {

A ^B C	Category	1.2 Average Profit
1	Furniture	9.456790123
2	Clothing	11.76290832
3	Electronics	34.07142857

Group By Use First Row as Headers Count Rows Rename Pivot Column Convert to List

Table Any Column

Queries [7]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Order Details Summary
- Avg Profit by Category
- Total Amount by SubCat...

fx = Table.AddColumn(#"Changed profit margin to", "Profit", {

A ^B C	Order ID	\$ Amount	Profit
1	B-25601	1,275.00	
2	B-25601	66.00	
3	B-25601	8.00	
4	B-25601	80.00	
5	B-25602	168.00	
6	B-25602	424.00	
7	B-25602	2,617.00	
8	B-25602	561.00	
9	B-25602	119.00	
10	B-25603	1,355.00	
11	B-25603	24.00	
12	B-25603	193.00	

Group By

Specify the column to group by and the desired output.

Basic Advanced

Sub-Category

New column name: Total Amount Operation: Sum Column: Amount

OK Cancel

Group By Use First Row as Headers Count Rows Rename Pivot Column Convert to List

Table Any Column

Queries [7]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Order Details Summary
- Avg Profit by Category
- Total Amount by SubCat...

fx = Table.Group(#"Added Conditional Column as Profit", "Sub-Category", {

A ^B C	Sub-Category	1.2 Total Amount
1	Bookcases	56861
2	Stole	18546
3	Hankerchief	14608
4	Electronic Games	39168
5	Phones	46119
6	Saree	53511
7	Trousers	30039
8	Chairs	34222
9	Kurti	3361
10	T-shirt	7382
11	Shirt	7555
12	Leggings	2106
13	Tables	22614
14	Printers	58252
15	Accessories	21728
16	Furnishings	13484
17	Skirt	1946

- Duplicate the Sales Target table and aggregate the total Target by Month of order date.

Duplicate Table

1. Right-click Sales Target
2. Select:

- Duplicate

Rename it:

- Sales Target Summary

Group By Month

1. Home → Group By
2. Configure:
 - Group By: Month
 - New column: Total Target
 - Operation: Sum
 - Column: Target
3. Click OK

Queries [8]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Order Details Summary
- Avg Profit by Category
- Total Amount by SubCat...
- Sales Target Summary**

fx = Table.TransformColumnTypes(#"Changed Type",{{"Targ

	Month of Order Date	Category	Target
1	01-04-2018	Furniture	
2	01-05-2018	Furniture	
3	01-06-2018	Furniture	
4	01-07-2018	Furniture	
5	01-08-2018	Furniture	
6	01-09-2018	Furniture	
7	01-10-2018	Furniture	
8	01-11-2018	Furniture	
9	01-12-2018	Furniture	
10	01-01-2019	Furniture	
11	01-02-2019	Furniture	
12	01-03-2019	Furniture	
13	01-04-2018	Clothing	
14	01-05-2018	Clothing	

fx = Table.RenameColumns(#"Inserted Month Name",{{"Month of Order Date", "Order Date"}, {"Month Name", "Month of Order Date"}})

Order Date	Category	Target	Month of Order Date
01-04-2018	Furniture	10,400.00	April
01-05-2018	Furniture	10,500.00	May
01-06-2018	Furniture	10,600.00	June
01-07-2018	Furniture	10,800.00	July
01-08-2018	Furniture	10,900.00	August
01-09-2018	Furniture	11,000.00	September
01-10-2018	Furniture	11,100.00	October
01-11-2018	Furniture		
01-12-2018	Furniture		
01-01-2019	Furniture		
01-02-2019	Furniture		
01-03-2019	Furniture		
01-04-2018	Clothing		
01-05-2018	Clothing		
01-06-2018	Clothing		
01-07-2018	Clothing		
01-08-2018	Clothing		
01-09-2018	Clothing		
01-10-2018	Clothing		
01-11-2018	Clothing		
01-12-2018	Clothing		
01-01-2019	Clothing		
01-02-2019	Clothing		
01-03-2019	Clothing	16,000.00	March
01-04-2018	Electronics	9,000.00	April
01-05-2018	Electronics	8,000.00	May

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Month of Order Date

New column name: Total Target Operation: Sum Column: Target

OK Cancel

Table		Any Column	
Queries [8]		= Table.Group(#"Renamed Columns", {"Month of O	
List of Orders		Month of Order Date	1.2 Total Target
Order Details		1 April	31400
Sales target		2 May	31500
Orders Data		3 June	31600
Order Details Summary		4 July	33800
Avg Profit by Category		5 August	33900
Total Amount by SubCat...		6 September	34000
Sales Target Summary		7 October	36100
		8 November	36300
		9 December	36400
		10 January	43500
		11 February	43600
		12 March	43800

10. Data Modeling:

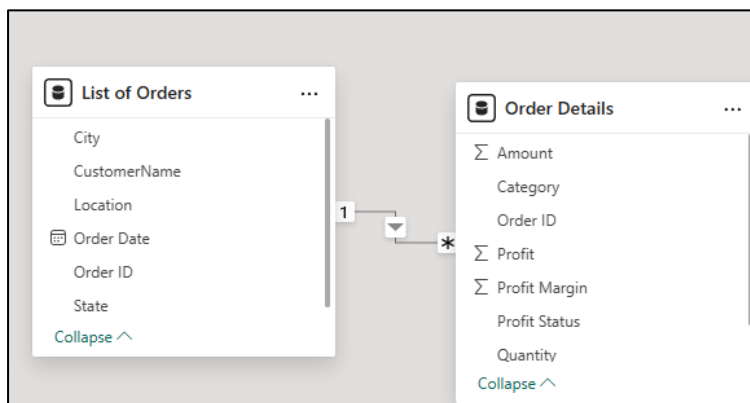
1. Establish a relationship between:

- List of Orders and Order Details using Order ID

In Power BI Desktop, click the Model View icon on the left sidebar

Steps

1. In Model View, locate both tables.
2. Drag:
 - Order ID from List of Orders
→ onto
 - Order ID in Order Details
3. A relationship window opens.



Establish a relationship between:

○ Order Details and Sales Target using Category

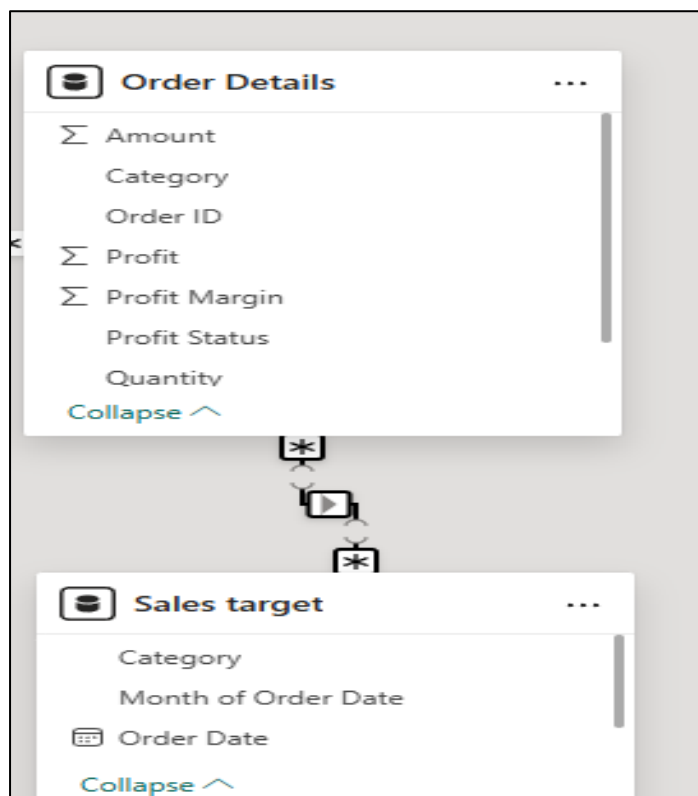
Steps

1. Drag:
 - Category from Order Details
→ onto
 - Category in Sales Target

2. Configure:

Setting	Value
Cardinality	Many-to-One or Many-to-Many
Cross filter direction	Single
Active	Checked

3. Click: OK



2. Use Manage Relationships to ensure relationships are active and correctly defined.

Use Manage Relationships

Go to:

Home → Manage Relationships

We could see both relationships listed.

Check:

- Status = Active
- Correct columns connected
- Correct tables connected

